Two samples against each other

Every feature of one sample plotted against another, with the correlation between them and the number of features it rests on.

AT A GLANCE

Where	Comparison → Pearson Correlation
Data levels	Protein, phosphosite, phosphoprotein
Needs	An uploaded dataset and its annotation, and two different samples or two different conditions.
Data source	Annotated data on the analysis scale, normalised if requested, but not completeness-filtered or imputed. Comparison has no Filtered or Imputed choice: filtering or imputing first would remove the disagreement being inspected.
Direction	The first selection is the x axis, the second the y axis. In the interactive plot the hovered difference is second minus first.
Two renderings	**Type** Normal draws a static figure; Interactive draws the same points in a pannable plot with hover and a fitted line.
Table	Yes, below the plot, downloadable as CSV.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

1 What the plot answers

A correlation matrix gives one number per pair of samples; this page gives the cloud behind one of those numbers. That is where the useful information is: whether the disagreement

between two samples is spread over the whole range or concentrated at the low end, whether it is a handful of features or all of them, and whether a high *r* is earned or merely a consequence of measuring intensities over four orders of magnitude.

Only features measured in *both* members of the pair are drawn. A feature present in one and missing in the other does not appear as a point on an axis; it is not on the figure at all, and the count *N* in the corner says how many features survived that test.

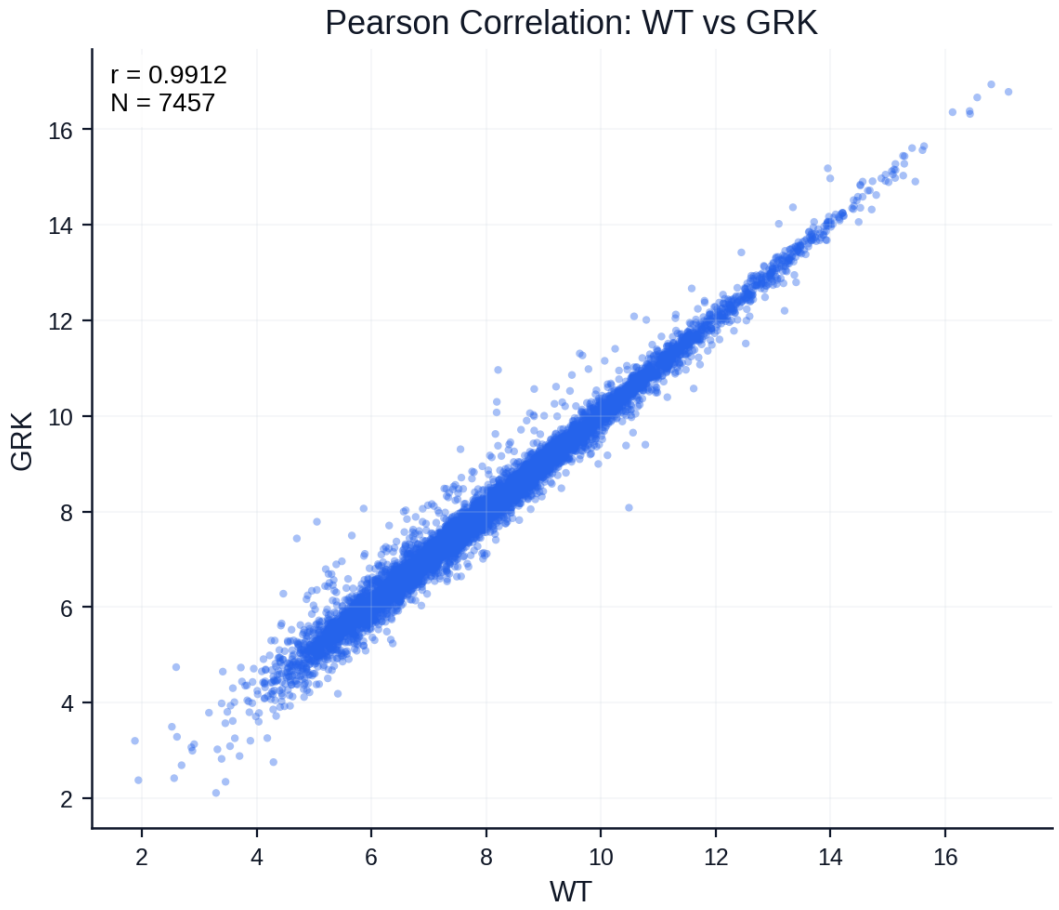


Figure 1.1. Mode Condition, **Type** Normal, protein level on the example project: the mean of the five WT samples on the x axis against the mean of the five GRK samples on the y. One point per protein, and the box in the corner gives the Pearson *r* and the number of features *N* it was computed on — here 0.9912 over the 7457 proteins the two conditions share. The cloud fans out below about $\log_2 6$ and is tight above it.

Reading it

- One point: one feature, at its $\log_2$ intensity in the first selection (x) and in the second (y).
- The box at the top left: r and N, the correlation and the number of features drawn.
- Points below the diagonal are lower in the second selection than in the first; points above it are higher.
- The fan at the low end: features near the detection limit, where the measurement is least reproducible. A widening of the cloud there is ordinary; a widening at the top is not.

- The axis labels are the sample or condition names, or the aliases if any were typed.

> **EASY TO GET WRONG**
>
> r is governed by the range of the data. Two samples that disagree about every regulated protein will still correlate above 0.98 if they agree about the four orders of magnitude separating the abundant proteins from the rare ones, so a high r is evidence that the measurement worked and almost no evidence about the biology. Read it together with N, which is not constant: each pair of samples is scored on the features that both of them measured, so a worse-covered sample is compared on fewer features and often scores *higher* for it.

2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which dataset to compare: Protein, Phospho or Phosphoprotein. Only the levels actually uploaded are offered.	Protein	page
Mode	Single compares two samples. Condition compares two condition means, each the mean over that condition's replicates.	Single	page
Type	Normal renders a static image, which is what the download button writes. Interactive renders a plot with hover, zoom and a fitted line, opened in its own window.	Normal	page
Sample 1 / Condition 1	The first member of the pair, drawn on the x axis. The label changes with **Mode**.	the first sample or condition of the annotation	page
Sample 2 / Condition 2	The second member, drawn on the y axis. It may not be the same as the first.	the first one that differs	page
Alias 1 (optional)	Replaces the first name on the x axis and in the title. The data are not affected.	empty	page
Alias 2 (optional)	The same for the second name and the y axis.	empty	page

Controls on this page (continued)

Control	What it does	Default	Where
Width, Height, DPI	The size of the *downloaded* figure. Width and height are held between 1 and 60 cm, the resolution between 72 and 600. Shown only while Normal is selected. The interactive plot has none: it fills its frame, Plotly's camera button saves a picture of it, and **Open Interactive Plot** opens it in a window of its own.	20 cm × 12 cm, 300	page
Download format	PNG, JPG, WEBP, SVG or PDF. Shown for Normal only.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Points	Colour, size, opacity, outline and shape of the dots. The figure uses its own values until a control here is touched: blue #1f77b4, about 7.7 pt across, 0.65 opaque, circular, with no outline.	as drawn	appearance
Reference line	The fitted line: shown or hidden, and its colour, style and width. Interactive only — the static figure draws no line, so this card has nothing to act on there.	shown, slate, solid, width 2	appearance
Title	Shown or hidden, and its text, size, weight, colour and position. The title is written as Pearson Correlation: first vs second.	shown	appearance
X axis, Y axis	The text and styling of the two axis labels and their tick labels.	the names or aliases	appearance
Frame and grid	Which of the four spines are drawn, and the grid behind the points. Normal opens with four spines and a solid grid on both axes; Interactive with no spines and its own pale grid.	by rendering	appearance
Font	One face for the whole figure, or one per text.	plot default	appearance
Background	Theme, paper colour and transparency.	Default, white, opaque	appearance

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here. **Dot Color** and **Dot Size**, which the

earlier manual listed beside the plot, are the **Points** card here.

3 How it is computed

HOW IT IS COMPUTED

Two vectors are taken from the $\log_2$ matrix, one per selection. In Single mode each is a sample column. In Condition mode each is the mean of that condition's replicates, taken over the replicates that have a value; a feature measured in one replicate of four gets a mean from that one replicate.
The pair is then reduced to the features present in both: a feature missing in either vector is dropped from both, and N is what remains. On those N pairs,

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2}\,\sqrt{\sum_i (y_i - \bar{y})^2}},$$

the product-moment correlation: the covariance of the two vectors over the product of their standard deviations, bounded by -1 and 1. It measures the straightness of the relationship, not its slope — a sample twice as intense as its replicate throughout correlates at 1.
r and N are printed in the corner of the figure. No p-value is computed here.

THE INTERACTIVE RENDERING

The same points, drawn with hover: each gives the feature identifier, its value in each selection, and the difference *second minus first*. A least-squares line $y = ax + b$ is fitted to the drawn points and extended over their range; the **Reference line** card hides it or restyles it. The feature identifier is taken from the dataset's own identifier column — protein names at protein level, the collapse key at phosphosite and phosphoprotein level — and is shortened in the hover box when it is long.

Edge cases

- The two selections must differ; the page refuses a pair of the same sample or the same condition rather than drawing a perfect diagonal.
- If no feature is present in both, no plot is drawn and the reason is stated.
- With a single surviving pair, r is undefined and the box reads r = n/a while the point is still drawn.

- A condition with one replicate is allowed: its mean is that replicate, and the axis label still says the condition name.
- N is specific to the pair on screen. Two comparisons from the same dataset are not scored on the same features, so their r values are not strictly comparable.
- Aliases are labels only. They change the axes, the title and the table, and nothing else.

Following up a difference

A narrow cloud indicates a strong linear relationship among the shared features, not necessarily equal measurements. For example, $y = x+1$ on a log2 plot can have $r = 1$ despite a twofold intensity offset. Inspect the cloud relative to equality as well as its correlation. Curvature, isolated points or changing spread are reasons to inspect those features and the preprocessing; they do not uniquely diagnose normalization failure. There is no universal pass mark such as $r > 0.95$. The overlap count and the experimental design matter alongside the coefficient.

4 The table below the plot

Two columns, Metric and Value, and four rows.

Rows in this table

Column	What it holds
Left	The first selection, as an alias if one was typed, otherwise the sample or condition name.
Right	The second selection, the same way.
Points	N: the features present in both, which is the number of dots on the figure.
PearsonR	r, to six decimals. Empty when it is undefined.

5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

```python
fig = correlation_xy_intensity(data, meta, "WT", "GRK")
```

```r
fig <- correlation_xy_intensity(data, meta, "WT", "GRK")
```

That is **Mode** Condition: the two names are conditions, and each axis is the mean over that condition's replicates. Single mode is a different function, correlation_xy_intensity_single, which takes two sample column names and no annotation at all. Both draw more than the static page does — a least-squares line is always fitted, and the header carries *r and* a p-value where the page prints *r* and *N* in a corner box and leaves the line to the interactive rendering. min_present has no control on the page: raise it above 1 to drop condition means that rest on fewer than that many replicates.

6 Exporting

With **Type** at Normal the download reproduces the figure at the width, height and resolution set on the page, which is why those three controls do not change what is on screen. With Interactive there is no image to download: the button becomes **Open Interactive Plot**, which opens the plot in its own window, and the plot's own toolbar saves a snapshot of the current view. The table underneath downloads separately as CSV in either case.

Behind this page

Function	What it produces
correlation_xy_intensity_single	Two samples, in a notebook. It also fits a regression line and prints a p-value, which the page does not.
correlation_xy_intensity	The same for two condition means
pearson_correlation_png	The static figure behind the page
pearson_correlation_html	The interactive figure, with the fitted line and the hover
pearson_correlation_table	The four-row table under the plot